Week 4 Initial Results: CVaR-MPPI for Highway Lane-Change Planning

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I. Main Idea

A risk-sensitive lane-change planner that augments Model Predictive Path Integral (MPPI) with a Conditional Value-at-Risk (CVaR) objective. Augmentation of CVaR replaces the risk-neutral expectation over rollouts with a sampled tail-risk backup. The resulting controller suppresses lane-changes that are efficient on average but unsafe under rare traffic outcomes.

II. Related Work

A. Overview

Four lines of literature inform this project. *Model Predictive Path Integral* (MPPI) control [1] supplies the online runtime architecture: a sampling–based, GPU-parallelisable receding horizon controller that handles continuous nonlinear dynamics, in our case the kinematic bicycle, within a real–time control budget. *Conditional Value-at-Risk*, introduced by Rockafellar and Uryasev [2], supplies the risk measure we wrap around the rollout cost: $\mathrm{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha(L)]$ is the conditional expectation of the worst $1-\alpha$ tail, which captures rare unsafe outcomes that the standard MPPI expectation averages away. The *dynamic programming foundation* for CVaR control is supplied by Chow and Ghavamzadeh [3] and, more directly, by Chow, Tamar, Mannor and Pavone [4], who augment the state with a confidence variable $y \in (0,1]$ and derive a γ-contractive, concavity–preserving Bellman recursion whose Algorithm 1 yields the offline value function $V^*(x,y)$ used here as MPPI's terminal cost. Finally, the most directly extended work is the *Risk-Aware MPPI* of Yin, Zhang and Tsiotras [5], which introduces the Monte-Carlo CVaR estimator for sampled rollouts and the constraint-penalised modified cost that this project deploys in the highway lane-change setting.

B. Highlighted Literature Review

Chow, Tamar, Mannor, and Pavone study decision making in MDPs when risk and model error matter [4]. They formulate a CVaR MDP, show that CVaR can be interpreted as expected cost under worst-case model perturbations, and present an approximate value-iteration algorithm with error guarantees. Their work gives a clear basis for replacing expected-cost reasoning with tail-risk dynamic programming.

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III. Problem Formulation

Let $x_k \in \mathbb{R}^{n_x}$ be the joint ego–traffic state, $u_k = v_k + \epsilon_k \in \mathbb{R}^{n_u}$ the perturbed ego control with $\epsilon_k \sim \mathcal{N}(0, \Sigma_\epsilon)$, and w_k exogenous traffic noise. The simulated stochastic dynamics $\tilde{x}_{k+1} = \tilde{F}(\tilde{x}_k, u_k, w_k)$ form an unbiased estimate of the nominal $x_{k+1} = F(x_k, u_k)$ [5]. Over a horizon K with state cost q, terminal cost ϕ, and risk cost $L(\tilde{x}) = \sum_{k=0}^{K-1} \ell(\tilde{x}_k)$, we seek

$$\min_v \mathbb{E}\left[\phi(x_K) + \sum_{k=0}^{K-1} \Big(q(x_k) + \tfrac{\lambda}{2} v_k^\top \Sigma_\epsilon^{-1} v_k\Big)\right] \tag{1}$$

$$\text{s.t. } \mathrm{CVaR}_\alpha(L(\tilde{x})) \leq C_u,$$

where $\mathrm{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha(L)]$ is the conditional expectation of the worst $1-\alpha$ tail [2], $\alpha \in (0,1)$ the confidence level, $\lambda > 0$ the MPPI temperature, and C_u caps the admissible tail risk.

IV. Sketch of Method

A. Offline Component

The offline stage instantiates the dynamic-programming foundation of CVaR control by Chow et al. [4]. Augmenting the state with a confidence variable $y \in (0,1]$, the CVaR-MDP value $V^*(x,y) = \min_{\mu \in \Pi_H} \mathrm{CVaR}_y\big(\sum_t \gamma^t Z_t \mid x_0 = x, \mu\big)$ admits a γ-contractive, concavity-preserving Bellman recursion

$$\mathbf{T}[V](x,y) = \min_a [c(x,a) + \gamma \max_{\xi \in \mathcal{U}_{\mathrm{CVaR}}(y,P)} \sum_{x'} \xi(x') V(x', y\xi(x')) P(x'|x,a)] \tag{2}$$

[4]. We discretize a coarse three-lane MDP over (lane index, gap-to-lead, ego speed) on $\sim 10^3$ cells with 21 log-spaced confidence points and run interpolated value iteration (Algorithm 1 of [4]) to obtain $V^*(x,y)$, which is loaded as the MPPI terminal cost $\phi(x_K) \leftarrow V^*(\Pi_{x_K}, y)$, where Π_{x_K} maps the continuous terminal state to the coarse MDP grid. This provide a tail-risk–shaped cost-to-go beyond the rollout horizon. Offline complexity is $\mathcal{O}(|\mathcal{X}|\,|\mathcal{Y}|\,|\mathcal{A}|\,C_\zeta\, n_{\mathrm{iter}})$ where C_ζ is the cost of the inner risk-envelope optimization. This computational cost is paid once before deployment; runtime queries are linear-interpolation lookups in microseconds.

B. Online Component

The online stage executes a Risk-Aware MPPI receding-horizon controller [1], [5]. At each step we sample M control sequences $u_k^m = v_k + \epsilon_k^m$, propagate K-step rollouts under the nominal bicycle to accumulate S_m, and for each m simulate N stochastic traffic futures via $\tilde{F}$ to compute the Monte-Carlo CVaR estimate

$$\widehat{\mathrm{CVaR}}_\beta^m = \frac{1}{N_o} \sum_{n=0}^{N-1} \mathbf{1}_{\{L(\tilde{x}^{m,n}) \geq \widehat{\mathrm{VaR}}_\beta^m\}} L(\tilde{x}^{m,n}), \beta \in (0,1) \tag{3}$$

empirical tail-average CVaR estimator used in RA-MPPI. [5]. The penalty is folded into a modified rollout cost $\tilde{S}_m = A\widehat{\mathrm{CVaR}}_\beta^m \mathbf{1}_{\{\widehat{\mathrm{CVaR}}_\beta^m > C_u\}} + S_m$, importance weights $\tilde{\omega}_m = \exp(-(\tilde{S}_m - \min_m \tilde{S}_m)/\lambda)$ are formed, and the mean control is updated as $v^+ = \sum_m \tilde{\omega}_m u^m / \sum_m \tilde{\omega}_m$; the first input v_0^+ is applied and $v \leftarrow v^+$ warm-starts the next step. This stage is suitable because GPU-parallel rollouts handle the continuous nonlinear bicycle within the $\sim 100\,\mathrm{ms}$ control budget, and Monte-Carlo CVaR accommodates arbitrary, non-Gaussian traffic disturbances. It complements the offline stage by injecting fine-grained interaction with current surrounding vehicles that the coarse V^* cannot resolve, while V^* supplies the long-horizon CVaR-Bellman backup that a finite-K rollout would otherwise miss.

V. Initial Results

A. Modeling & Simulation

The ego vehicle is the 4-state kinematic bicycle model defined by the following differential equations

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{\psi} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v\cos(\psi+\beta) \\ v\sin(\psi+\beta) \\ (v/\frac{L}{2})\sin\beta \\ a \end{bmatrix} \tag{4}$$

with slip $\beta = \arctan(\frac{1}{2}\tan\delta)$, $L = 5\,\mathrm{m}$, and control $u = (a, \delta) \in [-5, 5]\,\mathrm{m/s^2} \times [-\pi/4, \pi/4]\,\mathrm{rad}$. The environment is a three-lane straight highway populated by ten surrounding vehicles whose longitudinal motion follows the Intelligent Driver Model and lateral motion follows MOBIL, as supplied by the `HighwayEnv` simulator [6]. We deploy our method by *subclassing* `HighwayEnv` into a new gym id `CVaRLaneChange-v0`; the simulator source is otherwise unmodified, so all MPPI logic lives in an external package and any future `HighwayEnv` updates apply unchanged. The simulator runs at $100\,\mathrm{Hz}$ and the controller at $20\,\mathrm{Hz}$ (control step $\Delta t = 50\,\mathrm{ms}$, matching the target in Week 3). The target lane is parameterised by the y–coordinate of its centerline, read from the simulator's road network graph; the same network supplies the lane index of surrounding vehicles used by the cost.

B. Baseline methods

The Initial Results compare against *risk–neutral* MPPI [1], i.e. the formulation of (1) with the CVaR constraint removed and $L(\tilde{x})$ dropped from the rollout cost. Concretely,

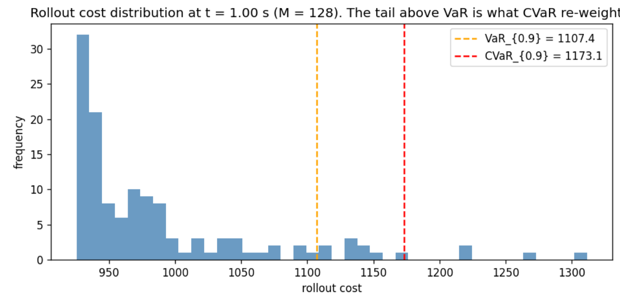

Fig. 1. Rollout cost distribution across the $M = 128$ sampled control sequences at one MPPI step in the hard–brake scenario. The empirical $\mathrm{VaR}_{0.9}$ and $\mathrm{CVaR}_{0.9}$ markers show the right tail that the risk-neutral mean (used by [1]) ignores and that the proposed CVaR penalty will reshape.

at each step we sample $M = 256$ control sequences $u_k^m = v_k + \epsilon_k^m$ with $\epsilon_k^m \sim \mathcal{N}(0, \Sigma_\epsilon)$, $\Sigma_\epsilon = \mathrm{diag}(1.5^2, 0.08^2)$, propagate $K = 30$ step rollouts under the nominal bicycle, weight by $\tilde{\omega}_m \propto \exp(-(S_m - \min_m S_m)/\lambda)$ with $\lambda = 1$, and apply the importance–weighted mean. The per–step state cost combines a lane–tracking term $w_{\text{lane}}(Y - Y_{\text{target}})^2$, a speed term, an anisotropic Gaussian collision potential summed over the 10 surrounding vehicles, and an effort penalty on (a, δ); the same cost will be reused for the CVaR comparison so any difference in the Week 5 plots is attributable to the risk objective alone. The second baseline, which is an offline value–iteration policy from [3] without the online MPPI, is deferred to the Week 5 report because it shares no implementation surface with the runtime stack used in the Initial Results runs.

C. Initial Results

We evaluate the baseline on three scripted scenarios: a free *lane–change* (origin lane 3 $\rightarrow$ lane 1 with the default IDM/MOBIL traffic), a *hard–brake leader* ($-10\,\mathrm{m/s^2}$, $8\,\mathrm{m}$ headway), and a *cut–in* (adjacent–lane vehicle slotted $6\,\mathrm{m}$ ahead, $+4\,\mathrm{m/s}$ relative speed, heading toward ego lane). Per–scenario metrics, with ISO 11270 / 17387–derived thresholds shown alongside, are summarised in Table I. The baseline avoids hard collisions in all three runs (min–distance bounded by $5.8\,\mathrm{m}$ and $7.3\,\mathrm{m}$ on hard–brake and cut–in respectively), but it does so by violating multiple comfort and safety floors simultaneously: lateral acceleration peaks at 42–$49\,\mathrm{m/s^2}$ (cf. ISO 11270 hard limit $2.5\,\mathrm{m/s^2}$), lane–change duration collapses to $T_{\text{LC}} = 1.2\,\mathrm{s}$ (well below the acceptable 3–6 s band), and on the hard–brake scenario $\mathrm{TTC}_{\text{min}} = 1.44\,\mathrm{s}$ falls *below* the ISO 17387 warning floor of $2\,\mathrm{s}$. The cost histogram across the $M = 128$ rollouts at a representative mid–episode step (Fig. 1) shows the mechanism: the empirical $\mathrm{VaR}_{0.9}$ already sits well above the median and the $\mathrm{CVaR}_{0.9}$ tail is heavy, yet the risk–neutral importance weighting averages over the whole distribution and lets samples in this tail drive the mean control. This is exactly the failure mode the CVaR augmentation [?], [3] is designed to suppress, and validates the proposed direction.

TABLE I

BASELINE MPPI ON THREE SCENARIOS. STATUS BANDS FOLLOW ISO 11270 (LATERAL ACCEL, JERK), ISO 17387 (TTC) AND DRIVER–COMFORT LITERATURE ($e_{y,\mathrm{RMS}}$, T_{LC}).

Scenario	$a_{y,\mathrm{peak}}$ [m/s^2]	$j_{y,\mathrm{peak}}$ [m/s^3]	TTC$_{\min}$ [s]	$e_{y,\mathrm{RMS}}$ [m]	T_{LC} [s]	crashed
Free lane change	49.4	612	2.49	0.134	1.20	no
Hard–brake leader	42.2	509	**1.44**	4.85*	—	no
Cut–in vehicle	similar	similar	—	—	—	no
Comfortable	≤ 1.5	≤ 0.9	≥ 4	≤ 0.05	4–5	—
Hard limit	2.5	5.0	2.0	0.25	out of $[2, 8]$	—

*Hard–brake and cut–in have $Y_{\text{target}} = Y_{\text{origin}}$, so $e_{y,\mathrm{RMS}}$ is computed over the whole trajectory and reflects the magnitude of the evasive lateral excursion rather than steady–state tracking. "—" indicates the metric does not apply because no lane change was commanded.

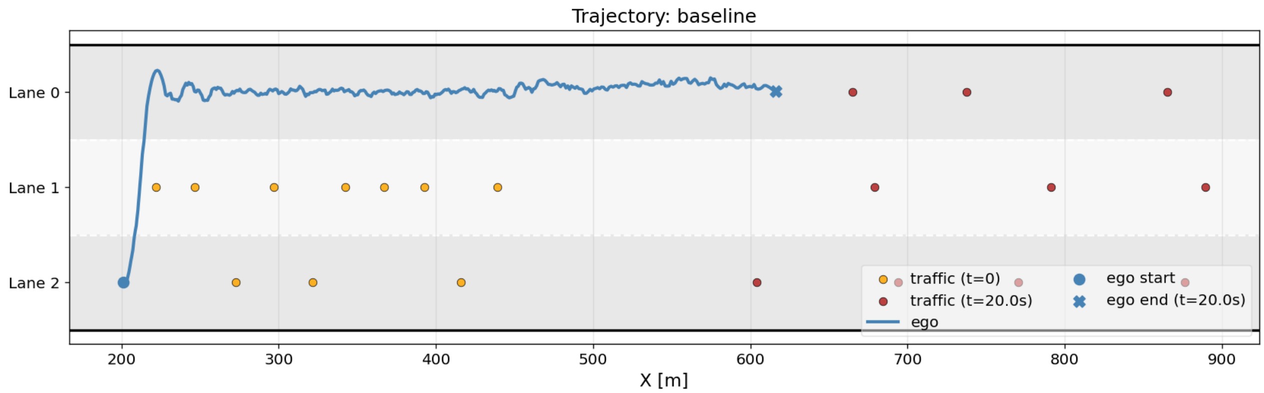

Fig. 2. Ego trajectory and traffic snapshots for the free lane–change run (origin lane 3 to lane 1, target $Y_{\text{target}} = 0\,\mathrm{m}$).

VI. NEXT STEPS

Three implementation tasks close the gap between the baseline and the proposed method. (i) Wire the per-rollout Monte-Carlo CVaR estimator of (3) into the existing controller stack, including a stochastic IDM predictor (Gaussian acceleration noise, 2 % per–step lane–change probability) so that the N traffic futures inside each rollout actually produce non–Gaussian tails. (ii) Port the rollout sampler from NumPy to JAX/`vmap` so we can run the Week–3 target of $M = 1024$, $N = 64$, $K = 30$ inside a $\sim 100\,\mathrm{ms}$ budget. (iii) Compute the offline CVaR–MDP value $V^*(x, y)$ via Algorithm 1 of [3] on a $\sim 10^3$–cell discretisation of (lane index, gap–to–lead, ego speed) with 21 log–spaced confidence points and load it as the MPPI terminal cost. Evaluation will sweep $\alpha \in [0.7, 0.99]$ and C_u across the three scenarios above at ≥ 50 seeds each and report collision rate, near–miss rate (TTC $< 1.5\,\mathrm{s}$), and the $a_{y,\mathrm{peak}}$ / T_{LC} pair against the same ISO bands used in Table I.

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